TECHKRITI'26

Team info

Team name: Entropy Vanguard Group

Team leader's name: Yash Lund (TK26-002952)

Contact details of team leader:
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Number of members in the team (max 6):1

MEMBERS: Yash Lund

GitHub repo : https://github.com/yashlund05/project_ari.git

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1) Introduction

Modern cryptocurrency markets are characterized by non-stationarity, where price dynamics shift rapidly between trending and sideways regimes. Traditional machine learning models often suffer from "regime drift," leading to significant drawdowns during market transitions. **Project ARI (Adaptive Regime Intelligence)** is a modular, Object-Oriented quantitative trading framework designed to overcome these challenges through unsupervised market state discovery and information-theoretic risk management.

The core innovation of Project ARI lies in its **Dual-Logic Regime Router**. Using a **Gaussian Hidden Markov Model (HMM)**, the system classifies the market into four distinct states: Bull, Bear, Sideways, and Recovery. Instead of utilizing a single "one-size-fits-all" model, ARI routes trade execution to specialized sub-engines. In Sideways and Bear regimes, the system employs a **LightGBM** mean-reversion engine to capture alpha from price oscillations. Conversely, in Bull and Recovery regimes, it utilizes a **Momentum-based Trend engine** to ride extended price expansions.

To ensure institutional-grade capital preservation, ARI integrates a **Shannon Entropy Kill-Switch**. By measuring the "uncertainty" in the HMM state probabilities, the system identifies periods of high market chaos. When entropy exceeds a threshold of 0.5, the system automatically triggers a **Hard Cash Lock**, bypassing all signals to protect equity.

In backtesting against 1,050 days of Bitcoin (BTC) out-of-sample data, Project ARI achieved a **Net Return of +49.51%**, generating an **Alpha of +27.16%** over the Buy & Hold benchmark. Crucially, the system maintained a **Max Drawdown of -24.26%**, successfully remaining above the competition's -30% penalty threshold. By incorporating a realistic **0.15% transaction fee**, ARI demonstrates a statistically robust and operationally viable path for automated cryptocurrency asset management.

2) References

- Competition Problem Statement:

  Techkriti IIT Kanpur – Beat the Market FinTech Challenge

- Dataset:

  Historical BTC/USDT market dataset provided by the competition organizers.

- Libraries and Tools:

  Python, Pandas, NumPy, Scikit-learn

- Concepts Referenced:

  Algorithmic Trading
  Quantitative Risk Management
  Machine Learning for Financial Time Series

- **GitHub Repository:** https://github.com/yashlund05/project_ari.git

- **Backtest Engine:** Custom Python Framework utilizing `backtester_V2_` and `hmmlearn`

3) Components Used

- **Core Model:** Gaussian Hidden Markov Model (HMM) for regime discovery.
- **ML Engine:** LightGBM (Gradient Boosting Machine) for mean-reversion signals.
- **Information Theory:** Shannon Entropy calculation for risk gating.
- **Software Stack:** Python 3.10+, Pandas, NumPy, Scikit-learn, Matplotlib.
- **Hardware:** Local Workstation with NVIDIA RTX 30-series GPU.
- Custom algorithmic trading strategy engine
- Custom backtesting system for strategy evaluation
- Historical BTC/USDT dataset (2020–2023)

4) ADDITIONAL INFORMATION

- Developed as a second-year AIML Engineering student based in Pune.
- Focused on implementing "Defensive Engineering" to survive the high-volatility environments typical of digital assets.
- This strategy achieved **+23.77% Equity Saved** purely through the entropy-based "Hard Cash Lock" logic during the 2022 market crash.

This project focuses on designing a modular and adaptive trading framework capable of responding to different cryptocurrency market conditions.

The modular design allows future integration of reinforcement learning approaches, ensemble models, and real-time trading simulations.